Анализ запусков (графики)

In [17]:
```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

Т.к. запуск тестов работает порядка 6 часов, использую информацию из `tests_output.md`

Запуск с checker ~ EXP

In [5]:
```python
all_n_df = pd.DataFrame([
    [2, 2, 1.0, 1.0, 3.986e-05, 1.734e-05],

    [3, 2, 1.0, 1.0, 3.129e-05, 1.768e-05],
    [3, 3, 1.0, 1.0, 2.718e-05, 1.922e-05],

    [4, 2, 1.0, 1.0, 8.6742e-04, 3.642e-05],
    [4, 3, 1.0, 1.0, 5.153e-05, 1.0275e-04],
    [4, 4, 1.0, 1.0, 6.638e-05, 1.2504e-03],

    [5, 2, 1.0, 1.0, 1.8774e-04, 2.7030e-04],
    [5, 3, 1.0294117647, 1.3333333333, 6.3620e-04, 2.57705e-03],
    [5, 4, 1.0058823529, 1.2, 1.6719e-04, 2.10215e-03],
    [5, 5, 1.0, 1.0, 1.1239e-04, 1.61852e-03],

    [6, 2, 1.0, 1.0, 9.506e-05, 2.3000e-04],
    [6, 3, 1.0541666667, 1.3333333333, 1.1698e-04, 2.03867e-03],
    [6, 4, 1.0530357143, 1.3333333333, 1.7999e-04, 7.03404e-03],
    [6, 5, 1.0119444444, 1.2, 2.2970e-04, 1.353693e-02],
    [6, 6, 1.0, 1.0, 1.0956e-04, 1.418687e-02],

    [7, 2, 1.0, 1.0, 7.244e-05, 3.0673e-04],
    [7, 3, 1.0066666667, 1.2, 1.0871e-04, 4.68572e-03],
    [7, 4, 1.0386904762, 1.3333333333, 1.2129e-04, 2.317264e-02],
    [7, 5, 1.0591798942, 1.2857142857, 1.3790e-04, 5.933948e-02],
    [7, 6, 1.0055555556, 1.1666666667, 1.4348e-04, 1.1373763e-01],
    [7, 7, 1.0, 1.0, 1.7071e-04, 2.9253281e-01],

    [8, 2, 1.0, 1.0, 1.3844e-04, 8.3213e-04],
    [8, 3, 1.0206349206, 1.1666666667, 1.6465e-04, 1.774221e-02],
    [8, 4, 1.0835714286, 1.25, 1.7402e-04, 1.3062159e-01],
    [8, 5, 1.0917378917, 1.25, 2.0098e-04, 5.5530618e-01],
    [8, 6, 1.0539245014, 1.1666666667, 1.9675e-04, 1.11556462],
    [8, 7, 1.0047619048, 1.0714285714, 2.2251e-04, 2.39140926],
    [8, 8, 1.0, 1.0, 2.2172e-04, 5.66769967],

    [9, 2, 1.0, 1.0, 1.5541e-04, 1.87737e-03],
    [9, 3, 1.05, 1.25, 1.9858e-04, 6.521212e-02],
    [9, 4, 1.075, 1.25, 2.1192e-04, 7.0658778e-01],
    [9, 5, 1.1449494949, 1.25, 2.6685e-04, 3.86383524],
    [9, 6, 1.0634199134, 1.1428571429, 2.6026e-04, 10.52491967],
    [9, 7, 1.0153846154, 1.0769230769, 2.7640e-04, 22.72462925],
    [9, 8, 1.0, 1.0, 2.5156e-04, 51.93782433],
    [9, 9, 1.0, 1.0, 3.2715e-04, 137.55043566],
], columns=[
    'n', 'k', 'avg_ratio', 'worst_ratio',
    'efficient_time', 'checker_time'
])
```

In [6]:
```python
connected_n_df = pd.DataFrame([
    [2, 2, 1.0, 1.0, 2.834e-05, 1.233e-05],

    [3, 2, 1.0, 1.0, 2.771e-05, 1.121e-05],
    [3, 3, 1.0, 1.0, 2.315e-05, 1.329e-05],

    [4, 2, 1.0, 1.0, 3.149e-05, 2.166e-05],
    [4, 3, 1.0, 1.0, 4.328e-05, 6.623e-05],
    [4, 4, 1.0, 1.0, 4.516e-05, 9.402e-05],

    [5, 2, 1.0, 1.0, 5.214e-05, 5.984e-05],
    [5, 3, 1.0476190476, 1.3333333333, 5.176e-05, 2.725e-04],
    [5, 4, 1.0095238095, 1.2, 5.192e-05, 5.7636e-04],
    [5, 5, 1.0, 1.0, 5.432e-05, 8.9637e-04],

    [6, 2, 1.0, 1.0, 5.917e-05, 1.3028e-04],
    [6, 3, 1.0366666667, 1.3333333333, 6.634e-05, 1.10963e-03],
    [6, 4, 1.0421428571, 1.3333333333, 8.593e-05, 3.96669e-03],
    [6, 5, 1.0083333333, 1.1666666667, 9.73e-05, 7.25321e-03],
    [6, 6, 1.0, 1.0, 1.166e-04, 1.376305e-02],
], columns=[
    'n', 'k', 'avg_ratio', 'worst_ratio',
    'efficient_time', 'checker_time'
])
```

In [7]:
```python
all_m_df = pd.DataFrame([
    [2, 2, 1.0, 1.0, 3.697e-05, 2.765e-05],

    [3, 2, 1.0, 1.0, 3.001e-05, 4.575e-05],
    [3, 3, 1.0, 1.0, 3.228e-05, 3.836e-04],

    [4, 2, 1.0, 1.0, 4.1e-05, 1.5699e-04],
    [4, 3, 1.0, 1.0, 5.341e-05, 2.60632e-03],
    [4, 4, 1.0, 1.0, 6.126e-05, 1.859014e-02],

    [5, 2, 1.0, 1.0, 5.711e-05, 3.0188e-04],
    [5, 3, 1.0138888889, 1.3333333333, 6.819e-05, 3.92468e-03],
    [5, 4, 1.0, 1.0, 7.54e-05, 2.595946e-02],
    [5, 5, 1.0, 1.0, 1.3219e-04, 1.5663263e-01],

    [6, 2, 1.0, 1.0, 9.74e-05, 5.6314e-04],
    [6, 3, 1.0333333333, 1.3333333333, 1.3729e-04, 1.066386e-02],
    [6, 4, 1.0, 1.0, 1.1485e-04, 5.50527e-02],
    [6, 5, 1.0, 1.0, 1.1375e-04, 1.3087975e-01],
    [6, 6, 1.0, 1.0, 1.2361e-04, 4.0590394e-01],
], columns=[
    'm', 'k', 'avg_ratio', 'worst_ratio',
    'efficient_time', 'checker_time'
])
```

In [8]:
```python
connected_m_df = pd.DataFrame([
    [2, 2, 1.0, 1.0, 4.043e-05, 1.94e-05],

    [3, 2, 1.0, 1.0, 2.834e-05, 1.826e-05],
    [3, 3, 1.0, 1.0, 2.706e-05, 5.73e-05],

    [4, 2, 1.0, 1.0, 3.275e-05, 4.128e-05],
    [4, 3, 1.0, 1.0, 3.385e-05, 1.9159e-04],
    [4, 4, 1.0, 1.0, 4.305e-05, 6.4923e-04],

    [5, 2, 1.0, 1.0, 4.261e-05, 8.686e-05],
    [5, 3, 1.0277777778, 1.3333333333, 4.522e-05, 6.8516e-04],
    [5, 4, 1.0, 1.0, 4.945e-05, 2.35746e-03],
    [5, 5, 1.0, 1.0, 5.649e-05, 7.19549e-03],

    [6, 2, 1.0, 1.0, 4.824e-05, 1.8263e-04],
    [6, 3, 1.05, 1.3333333333, 5.128e-05, 2.10325e-03],
    [6, 4, 1.0131578947, 1.25, 8.332e-05, 1.108433e-02],
    [6, 5, 1.0, 1.0, 1.0114e-04, 3.360607e-02],
    [6, 6, 1.0, 1.0, 1.2145e-04, 1.1448273e-01],
], columns=[
    'm', 'k', 'avg_ratio', 'worst_ratio',
    'efficient_time', 'checker_time'
])
```

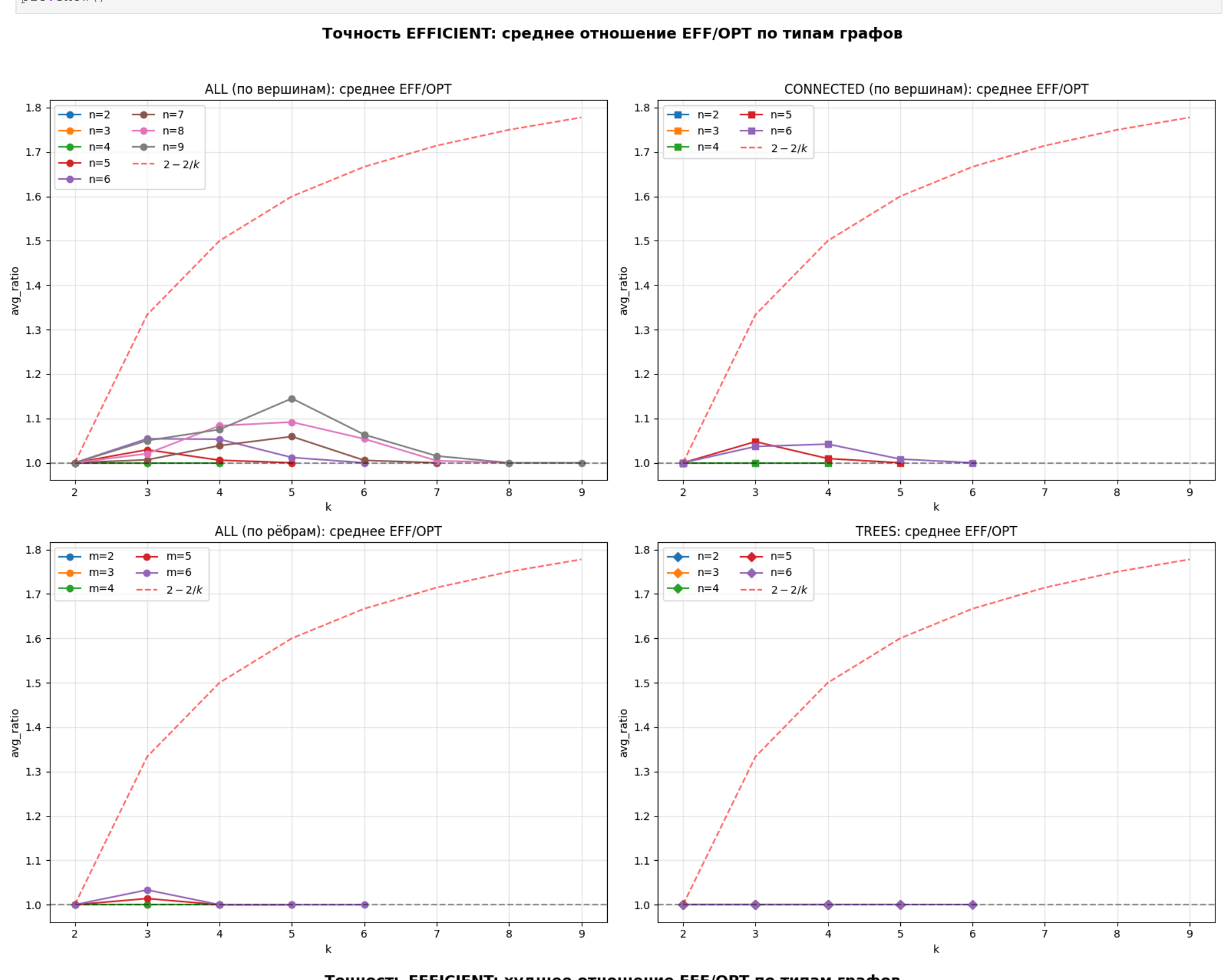
In [9]:
```python
trees_df = pd.DataFrame([
    [2, 2, 1.0, 1.0, 8.95e-06, 4.28e-06],

    [3, 2, 1.0, 1.0, 1.409e-05, 7.66e-06],
    [3, 3, 1.0, 1.0, 1.617e-05, 1.11e-05],

    [4, 2, 1.0, 1.0, 2.068e-05, 1.93e-05],
    [4, 3, 1.0, 1.0, 2.305e-05, 5.563e-05],
    [4, 4, 1.0, 1.0, 2.648e-05, 8.116e-05],

    [5, 2, 1.0, 1.0, 2.841e-05, 5.029e-05],
    [5, 3, 1.0, 1.0, 3.222e-05, 2.6301e-04],
    [5, 4, 1.0, 1.0, 3.922e-05, 5.843e-04],
    [5, 5, 1.0, 1.0, 4.031e-05, 9.1393e-04],

    [6, 2, 1.0, 1.0, 3.744e-05, 1.2504e-04],
    [6, 3, 1.0, 1.0, 4.436e-05, 1.12703e-03],
    [6, 4, 1.0, 1.0, 6.893e-05, 4.30549e-03],
    [6, 5, 1.0, 1.0, 7.198e-05, 7.57781e-03],
    [6, 6, 1.0, 1.0, 8.947e-05, 1.390351e-02],
], columns=[
    'n', 'k', 'avg_ratio', 'worst_ratio',
    'efficient_time', 'checker_time'
])
```

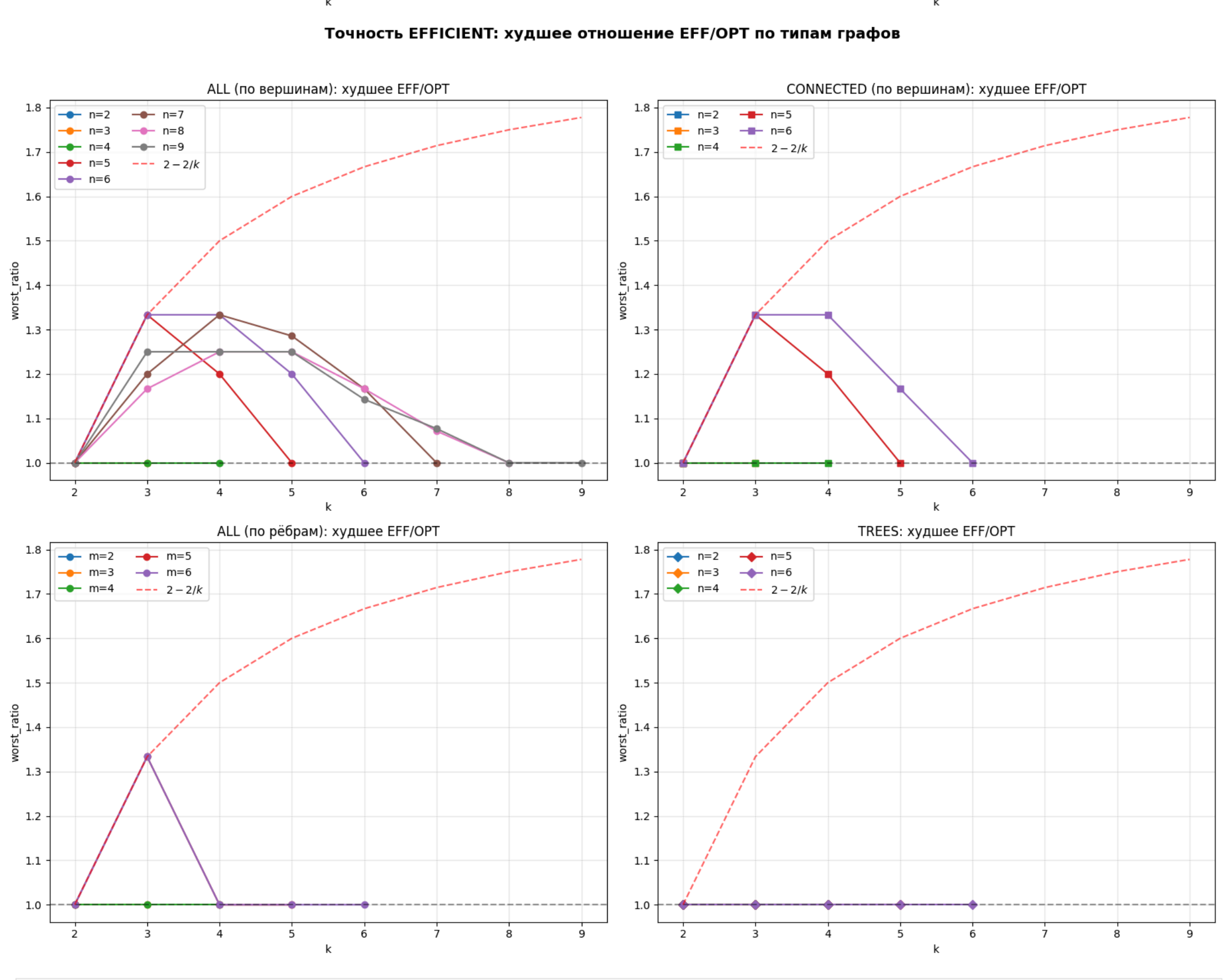
Запуск только EFFICIENT (замер времени):

In [15]:
```python
all_time_data = [
    # m, k, time
    (2,2,4.875e-05),

    (3,2,2.893e-05),(3,3,2.5e-05),

    (4,2,3.441e-05),(4,3,3.115e-05),(4,4,3.3e-05),

    (5,2,3.578e-05),(5,3,3.634e-05),(5,4,3.943e-05),(5,5,4.208e-05),

    (6,2,4.457e-05),(6,3,4.575e-05),(6,4,4.944e-05),
    (6,5,9.505e-05),(6,6,7.316e-05),

    (7,2,5.084e-05),(7,3,5.44e-05),(7,4,5.83e-05),
    (7,5,7.07e-05),(7,6,9.761e-05),(7,7,0.00010072),

    (8,2,6.225e-05),(8,3,6.147e-05),(8,4,6.519e-05),
    (8,5,7.096e-05),(8,6,8.081e-05),(8,7,7.851e-05),(8,8,8.142e-05),

    (9,2,7.052e-05),(9,3,7.245e-05),(9,4,7.575e-05),
    (9,5,8.031e-05),(9,6,8.463e-05),(9,7,8.968e-05),
    (9,8,9.332e-05),(9,9,9.756e-05),

    (10,2,7.829e-05),(10,3,7.971e-05),(10,4,8.431e-05),
    (10,5,8.982e-05),(10,6,9.261e-05),(10,7,9.682e-05),
    (10,8,0.00010213),(10,9,0.00010625),(10,10,0.00010827),

    (11,2,9.987e-05),(11,3,9.36e-05),(11,4,9.702e-05),
    (11,5,0.00010175),(11,6,0.00010655),(11,7,0.00011199),
    (11,8,0.00011804),(11,9,0.00012274),(11,10,0.0001279),
    (11,11,0.00013154),

    (12,2,9.406e-05),(12,3,9.514e-05),(12,4,9.924e-05),
    (12,5,0.00010452),(12,6,0.00011056),(12,7,0.00011366),
    (12,8,0.00012109),(12,9,0.00012565),(12,10,0.00013052),
    (12,11,0.00013432),(12,12,0.00013593),

    (13,2,0.00011646),(13,3,0.00010872),(13,4,0.00011367),
    (13,5,0.00011792),(13,6,0.00013423),(13,7,0.00013595),
    (13,8,0.00013634),(13,9,0.00014509),(13,10,0.00015069),
    (13,11,0.00015221),(13,12,0.00015459),(13,13,0.00015674),

    (14,2,0.00012165),(14,3,0.00014251),(14,4,0.00012909),
    (14,5,0.00014353),(14,6,0.00013963),(14,7,0.00014581),
    (14,8,0.00015054),(14,9,0.00015756),(14,10,0.00016853),
    (14,11,0.00016904),(14,12,0.00017356),(14,13,0.00017986),
    (14,14,0.00018456),

    (15,2,0.00013598),(15,3,0.00012826),(15,4,0.00013187),
    (15,5,0.00013747),(15,6,0.00014342),(15,7,0.00014973),
    (15,8,0.00015611),(15,9,0.00016536),(15,10,0.00016858),
    (15,11,0.00017571),(15,12,0.00018164),(15,13,0.00018387),
    (15,14,0.00018057),(15,15,0.00018323),

    (16,2,0.00014041),(16,3,0.00013568),(16,4,0.00013683),
    (16,5,0.0001423),(16,6,0.00014745),(16,7,0.00015228),
    (16,8,0.00015976),(16,9,0.0001676),(16,10,0.00017235),
    (16,11,0.00017829),(16,12,0.00018057),(16,13,0.00018109),
    (16,14,0.00019155),(16,15,0.00018951),(16,16,0.00020066),
]

connected_time_data = [
    (2,2,4.924e-05),

    (3,2,3.374e-05),(3,3,2.658e-05),

    (4,2,3.611e-05),(4,3,3.121e-05),(4,4,3.333e-05),

    (5,2,3.855e-05),(5,3,3.987e-05),
    (5,4,4.075e-05),(5,5,4.302e-05),

    (6,2,4.01e-05),(6,3,4.038e-05),(6,4,4.263e-05),
    (6,5,4.57e-05),(6,6,4.867e-05),

    (7,2,5.577e-05),(7,3,6.014e-05),(7,4,7.134e-05),
    (7,5,8.741e-05),(7,6,0.00010685),(7,7,0.00011399),

    (8,2,0.00010309),(8,3,0.00010015),(8,4,9.139e-05),
    (8,5,8.791e-05),(8,6,9.016e-05),(8,7,9.488e-05),(8,8,9.588e-05),

    (9,2,6.684e-05),(9,3,6.915e-05),(9,4,7.152e-05),
    (9,5,7.715e-05),(9,6,8.047e-05),(9,7,8.425e-05),
    (9,8,8.837e-05),(9,9,9.053e-05),

    (10,2,7.094e-05),(10,3,7.088e-05),(10,4,7.682e-05),
    (10,5,8.43e-05),(10,6,8.817e-05),(10,7,9.224e-05),
    (10,8,9.725e-05),(10,9,0.00010155),(10,10,0.0001035),

    (11,2,8.08e-05),(11,3,8.041e-05),(11,4,9.071e-05),
    (11,5,9.548e-05),(11,6,0.0001022),(11,7,0.00010909),
    (11,8,0.00011242),(11,9,0.00011569),(11,10,0.00011846),
    (11,11,0.0001212),

    (12,2,9.327e-05),(12,3,9.186e-05),(12,4,9.705e-05),
    (12,5,0.00010428),(12,6,0.00010931),(12,7,0.00011425),
    (12,8,0.00011895),(12,9,0.00012454),(12,10,0.00012937),
    (12,11,0.00013155),(12,12,0.00013276),

    (13,2,0.00010756),(13,3,0.0001153),(13,4,0.00011735),
    (13,5,0.00012211),(13,6,0.00012827),(13,7,0.000135),
    (13,8,0.00013948),(13,9,0.00014478),(13,10,0.00015059),
    (13,11,0.00015704),(13,12,0.00016158),(13,13,0.00016244),

    (14,2,0.00010686),(14,3,0.00011424),(14,4,0.0001163),
    (14,5,0.00012189),(14,6,0.00012784),(14,7,0.00013407),
    (14,8,0.00013981),(14,9,0.00014652),(14,10,0.00014816),
    (14,11,0.00014764),(14,12,0.00014975),(14,13,0.00014972),
    (14,14,0.00014878),

    (15,2,0.00011309),(15,3,0.00021777),(15,4,0.00015002),
    (15,5,0.00012604),(15,6,0.00013107),(15,7,0.00013709),
    (15,8,0.00014327),(15,9,0.00014953),(15,10,0.00015276),
    (15,11,0.00015335),(15,12,0.00015384),(15,13,0.00015284),
    (15,14,0.00015305),(15,15,0.00015315),

    (16,2,0.00012133),(16,3,0.00012779),(16,4,0.00012995),
    (16,5,0.00013521),(16,6,0.0001405),(16,7,0.00014664),
    (16,8,0.0001529),(16,9,0.00015984),(16,10,0.00016386),
    (16,11,0.00016441),(16,12,0.00016595),(16,13,0.00016517),
    (16,14,0.00016499),(16,15,0.00016447),(16,16,0.00016496),
]
```

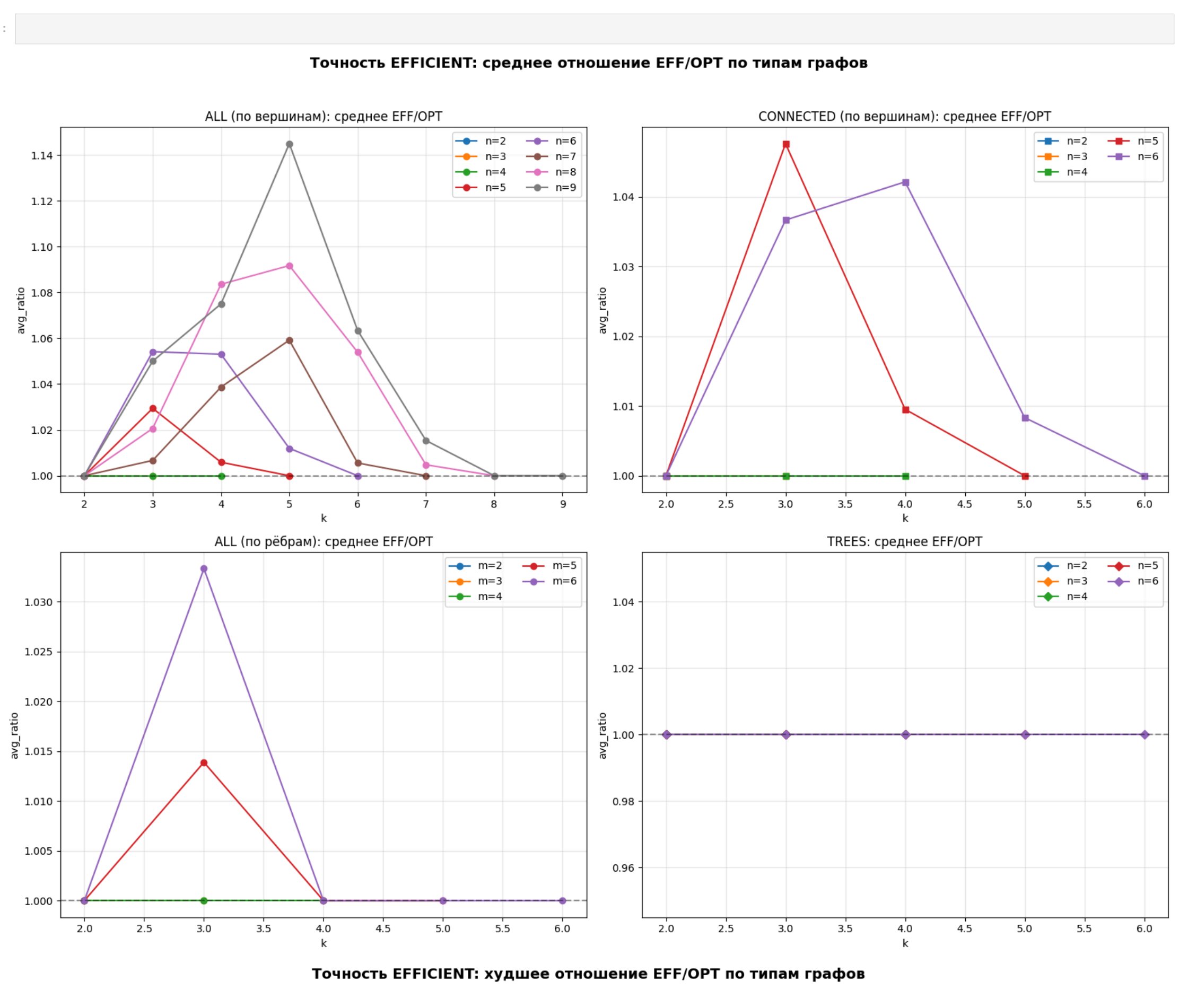
Графики

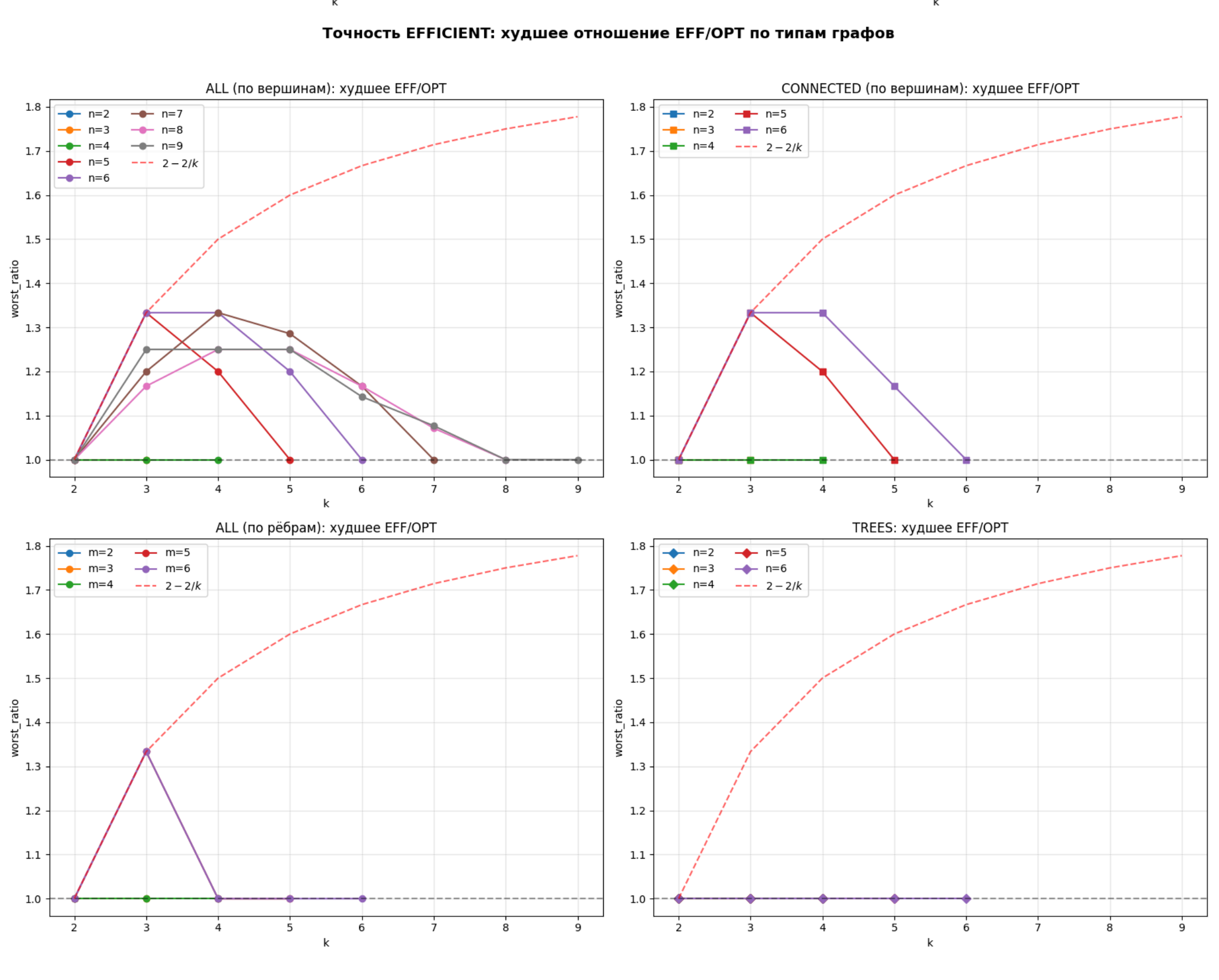
In [24]:
```python
''' Средняя точность EFFICIENT '''
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

k_vals = np.array(sorted(all_n_df["k"].unique()))
bound = 2 - 2/k_vals

# --- ALL (по числу вершин n) ---
for n_val in sorted(all_n_df["n"].unique()):
    subset = all_n_df[all_n_df["n"] == n_val]
    axes[0, 0].plot(
        subset["k"],
        subset["avg_ratio"],
        marker='o',
        label=f"n={n_val}"
    )

axes[0, 0].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes[0, 0].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes[0, 0].set_title("ALL (по вершинам): среднее EFF/OPT")
axes[0, 0].set_xlabel("k")
axes[0, 0].set_ylabel("avg_ratio")
axes[0, 0].grid(True, alpha=0.3)
axes[0, 0].legend(ncol=2)

# --- CONNECTED (по числу вершин n) ---
for n_val in sorted(connected_n_df["n"].unique()):
    subset = connected_n_df[connected_n_df["n"] == n_val]
    axes[0, 1].plot(
        subset["k"],
        subset["avg_ratio"],
        marker='s',
        label=f"n={n_val}"
    )

axes[0, 1].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes[0, 1].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes[0, 1].set_title("CONNECTED (по вершинам): среднее EFF/OPT")
axes[0, 1].set_xlabel("k")
axes[0, 1].set_ylabel("avg_ratio")
axes[0, 1].grid(True, alpha=0.3)
axes[0, 1].legend(ncol=2)

# --- ALL (по числу рёбер m) ---
for m_val in sorted(all_m_df["m"].unique()):
    subset = all_m_df[all_m_df["m"] == m_val]
    axes[1, 0].plot(
        subset["k"],
        subset["avg_ratio"],
        marker='o',
        label=f"m={m_val}"
    )

axes[1, 0].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes[1, 0].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes[1, 0].set_title("ALL (по рёбрам): среднее EFF/OPT")
axes[1, 0].set_xlabel("k")
axes[1, 0].set_ylabel("avg_ratio")
axes[1, 0].grid(True, alpha=0.3)
axes[1, 0].legend(ncol=2)

# --- TREES ---
for n_val in sorted(trees_df["n"].unique()):
    subset = trees_df[trees_df["n"] == n_val]
    axes[1, 1].plot(
        subset["k"],
        subset["avg_ratio"],
        marker='D',
        label=f"n={n_val}"
    )

axes[1, 1].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes[1, 1].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes[1, 1].set_title("TREES: среднее EFF/OPT")
axes[1, 1].set_xlabel("k")
axes[1, 1].set_ylabel("avg_ratio")
axes[1, 1].grid(True, alpha=0.3)
axes[1, 1].legend(ncol=2)

fig.suptitle("Точность EFFICIENT: среднее отношение EFF/OPT по типам графов",
             fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()


''' Худшее отношение'''
fig2, axes2 = plt.subplots(2, 2, figsize=(16, 12))

# ALL (n)
for n_val in sorted(all_n_df["n"].unique()):
    subset = all_n_df[all_n_df["n"] == n_val]
    axes2[0, 0].plot(
        subset["k"],
        subset["worst_ratio"],
        marker='o',
        label=f"n={n_val}"
    )

axes2[0, 0].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes2[0, 0].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes2[0, 0].set_title("ALL (по вершинам): худшее EFF/OPT")
axes2[0, 0].set_xlabel("k")
axes2[0, 0].set_ylabel("worst_ratio")
axes2[0, 0].grid(True, alpha=0.3)
axes2[0, 0].legend(ncol=2)

# CONNECTED (n)
for n_val in sorted(connected_n_df["n"].unique()):
    subset = connected_n_df[connected_n_df["n"] == n_val]
    axes2[0, 1].plot(
        subset["k"],
        subset["worst_ratio"],
        marker='s',
        label=f"n={n_val}"
    )

axes2[0, 1].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes2[0, 1].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes2[0, 1].set_title("CONNECTED (по вершинам): худшее EFF/OPT")
axes2[0, 1].set_xlabel("k")
axes2[0, 1].set_ylabel("worst_ratio")
axes2[0, 1].grid(True, alpha=0.3)
axes2[0, 1].legend(ncol=2)

# ALL (m)
for m_val in sorted(all_m_df["m"].unique()):
    subset = all_m_df[all_m_df["m"] == m_val]
    axes2[1, 0].plot(
        subset["k"],
        subset["worst_ratio"],
        marker='o',
        label=f"m={m_val}"
    )

axes2[1, 0].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes2[1, 0].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes2[1, 0].set_title("ALL (по рёбрам): худшее EFF/OPT")
axes2[1, 0].set_xlabel("k")
axes2[1, 0].set_ylabel("worst_ratio")
axes2[1, 0].grid(True, alpha=0.3)
axes2[1, 0].legend(ncol=2)

# TREES
for n_val in sorted(trees_df["n"].unique()):
    subset = trees_df[trees_df["n"] == n_val]
    axes2[1, 1].plot(
        subset["k"],
        subset["worst_ratio"],
        marker='D',
        label=f"n={n_val}"
    )

axes2[1, 1].axhline(y=1.0, color='black', linestyle='--', alpha=0.4)
axes2[1, 1].plot(k_vals, bound, 'r--', linewidth=1.5, alpha=0.6, label='$2-2/k$')
axes2[1, 1].set_title("TREES: худшее EFF/OPT")
axes2[1, 1].set_xlabel("k")
axes2[1, 1].set_ylabel("worst_ratio")
axes2[1, 1].grid(True, alpha=0.3)
axes2[1, 1].legend(ncol=2)

fig2.suptitle("Точность EFFICIENT: худшее отношение EFF/OPT по типам графов",
              fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```





In [ ]:





Выводы

1. **Оценка $(2 - \frac{2}{k})$ не нарушена.**
   На всех протестированных графах отношение `EFF/OPT` не превысило теоретическую верхнюю границу.

2. **Точность на практике:**

Класс графов	Среднее EFF/OPT	Худшее `EFF/OPT`
Все графы	1.0 – 1.14	1.33 (k=3)
Связные графы	1.0 – 1.10	1.33
По числу рёбер	1.0 – 1.11	1.33
Деревья	**всегда 1.0**	**всегда 1.0**

На деревьях EFFICIENT находит **минимальный** k-разрез, потому что в дереве каждый разрез — удаление одного ребра, и жадный алгоритм совпадает с оптимальным.

3. **Граница $(2 - \frac{2}{k})$ сильно завышена для реальных графов.**
   Даже худшее отношение 1.33 значительно ниже теоретического *max_bound* = $2 - \frac{2}{k}$. Графы, на которых bound достигается, — искусственные конструкции (см [SV]).

4. **Пик ошибки — при $(k \approx n/2)$.**

5. **EFFICIENT всегда точен при $k = 2$ и $k = n$.**
   Для $k = 2$ дерево Гомори-Ху находит глобальный минимальный разрез. Для k=n разрез единственен — удаление всех рёбер.

6. **Выигрыш по времени.**
   EFFICIENT для n $\leq$ 16 работает за десятки микросекунд (несмотря на ассимпотику $O(|V|^2|E|^2)$, из графиков видно, что алгоритм работает почти линейно по E). Оптимальный алгоритм уже при $n = 9$ требует нескольких минут.